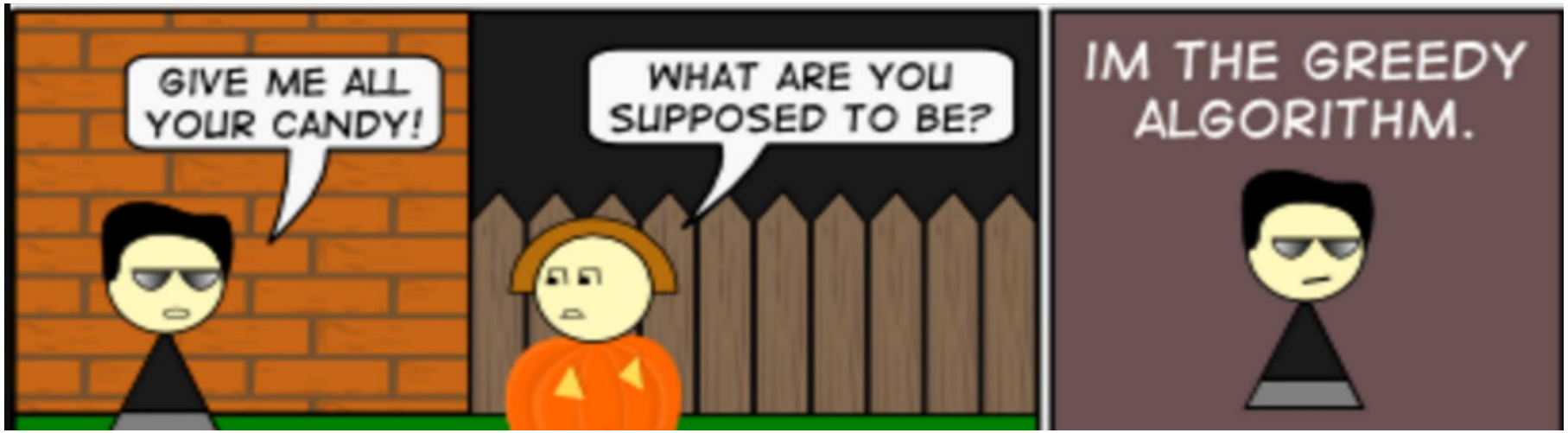
Algorithm Design : Greedy Algorithms

The seven deadly sins in Christianity (Also in other religions) :

Envy, gluttony, greed , lust, pride, sloth, wrath



Optimization problems

- An optimization problem is one in which you want to find, not just *a* solution, but the *best* solution
- A "greedy algorithm" sometimes works well for most optimization problems
- A greedy algorithm works in phases. At each phase:
  - You take the best you can get right now, without regard for future consequences
  - You hope that by choosing a *local* optimum at each step, you will end up at *global* optimum

In order to get what you want, just grab what looks best!

Greedy Properties

1. "greedy-choice property" It says that a globally optimal solution can be arrived at by making a series of locally optimal choices.
2. "optimal substructure" A problem exhibits optimal substructure if an optimal solution to the problem contains optimal solutions to the sub-problems.

→In order for greedy heuristic to solve the problem, the optimal solution to the big problem should contain optimal solutions to sub problems

The Structure of Greedy Algorithms

```
Algorithm Greedy (a , n)
//a[1:n]contains the n inputs.
   solution = 0    //initialize the solution.
   for i =1 to n do
      {
            x =Select(a)
            if Feasible( solution, x) then
               solution = Union(solution,x)  //Accept partial sol.
      }
   return solution
```

- *Select*() selects an input from a[] and removes it.

  the selected input value is assigned to x.
- *Feasible*() is a boolean-valued function that determines whether x can be included into the solution vector (no constraints are violated?).
- Union() combines x with the solution so far and updates the objective function.

Example : Coin Changing Problem

- Greedy Algorithm works by making the decision that seems most promising at any moment; it never reconsiders this decision, whatever situation may arise later.
- As an example consider the problem of "Making Change".
- Assume coins available are:
  - 100 cents,25 cents,10 cents ,5 cents,1 cent
- A Greedy Algorithm : Use fewest possible bills and coins

→At each step, take the largest possible bill or coin that does not make the sum > required

Coin changing problem

//Goal: Make change for n units using the least possible number of coins.

MAKE-CHANGE (n)

```
C ← {100, 25, 10, 5, 1}     // constant.
S← {};                      // Initially empty set that will hold the solutions
Sum ← 0 sum of items in solution set
WHILE sum != n
   x = largest item in set C such that (sum + x ) ≤ n  //Greedy choice
   IF no such item THEN
       RETURN    "No Solution“     //Not feasible
   S ← S Union (value of x)        //Include x to solution set
   sum ← sum + x
RETURN S
```

How the Algorithm Works

- **Example :** Make a change for 2.89 (289 cents) here n = 2.89 and the solution contains 2 dollars, 3 quarters, 1 dime and 4 pennies.

  (A dollar = 100 cents, a quarter =25 cents, a dime =10 cents, a penny=1 cent)

- The algorithm is greedy because at every stage it chooses the largest coin without worrying about the consequences.
- Moreover, it never changes its choice : once a coin has been included in the solution set, it remains there.

Note:The greedy solution may not work for some currency systems.

Some Greedy Algorithms

- Dijkstra's algorithm for finding single source shortest paths in a graph
- Kruskal's algorithm for finding a minimum-cost spanning tree
- Prim's algorithm for finding a minimum-cost spanning tree
- Huffman algorithm for finding minimal length codes for data compression
- Knapsack algorithm for filling optimally a space with different items

Single Source Weighted Shortest Paths

- The problem: Given a graph $G=(V, E, W)$ for which there is no negative weight and one of the vertices is specified to be the source vertex.
- Determine the cost of the shortest paths from the source to every other vertex in V.

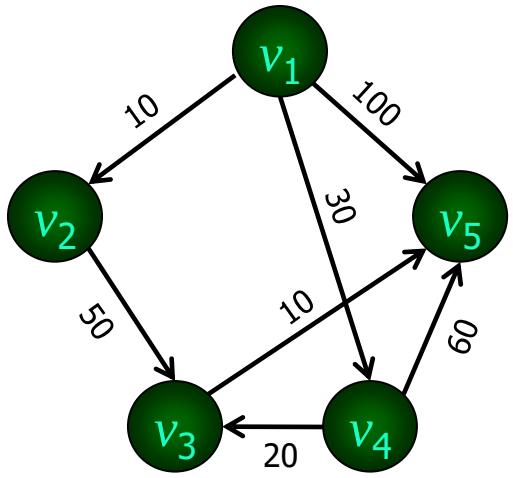
Dijkstra's Algorithm can be used to solve this problem.

(Developed by E.W.Dijkstra,1959)

An Application of Weighted Shortest Paths

The graph G represents an airline map.

- Vertices in G represent cities
- Each directed edge in G represents a route from one city to another city.
- Each weight represents the time required for flying from one city to another city.
- **When we solve the single source shortest path problem, we will be able to determine the minimum flight time from a given city (v1) to every other city on the map.**



Dijkstra's Algorithm

- Dijkstra's algorithm – A greedy solution to the single-source shortest paths problem in graph theory.

- **Works on both directed and undirected graphs. However, all edges must have nonnegative weights.**

Input: Weighted graph G={E,V,W} and source vertex $s \in V$.

Output: Lengths of shortest paths (or the shortest paths themselves) **from a given source vertex $s \in V$ to all other vertices**

Dijkstra's Algorithm : Idea

- Problem:We have a weighted graph G= (V,E,W).

  Find shortest paths from a given node to all other nodes.
- Start with source vertex s and iteratively construct a tree rooted at s
- Each vertex keeps track of current cheapest path from s
- At each iteration, include the vertex whose cheapest path from s is the overall cheapest.

→ The choice is greedy !

Dijkstra's Algorithm - Pseudocode

```
//Weighted graph G and source vertex  s are available
Dijkstra(G,s)
dist[s] ← 0                                  //distance to source vertex is zero
for  all v ∈ V–{s}
      do  dist[v] ←∞                         //set all other distances to infinity
S←∅                                          //S, visited vertices set. Initially empty
Q←V                                          //Q, the queue initially contains all vertices)
while Q ≠∅                                   //while the queue is not empty
     do   u ← mindistance(Q, dist[v])//select the el. of Q with the min. dist. from s
     S←S∪{u}                                 //add u to list of visited vertices
      for all v ∈ neighbors[u]
           do  if   dist[u] + w(u, v) < dist[v]        // if new shortest path found
                    then      d[v] ←d[u] + w(u, v)   //Update shortest path
            // ..................... (if desired, add code here to display the path
return dist                                  //dist now includes shortest distances from s
end   //Dijkstra
```

Example : Initialization

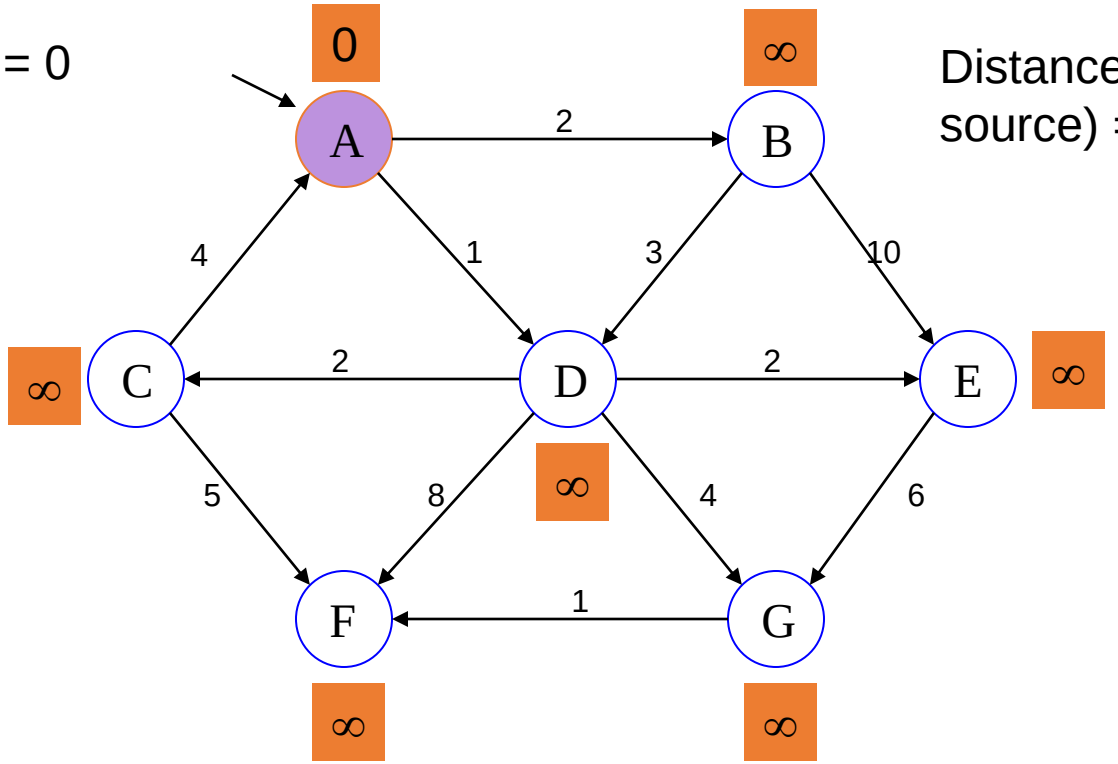
Q ={A,B,C,D,E,F,G} , source A, S={A}, A is known(visited)

Distance(source) = 0

Distance(D) = 1

Distance(B) = 2

Distance(D)=4

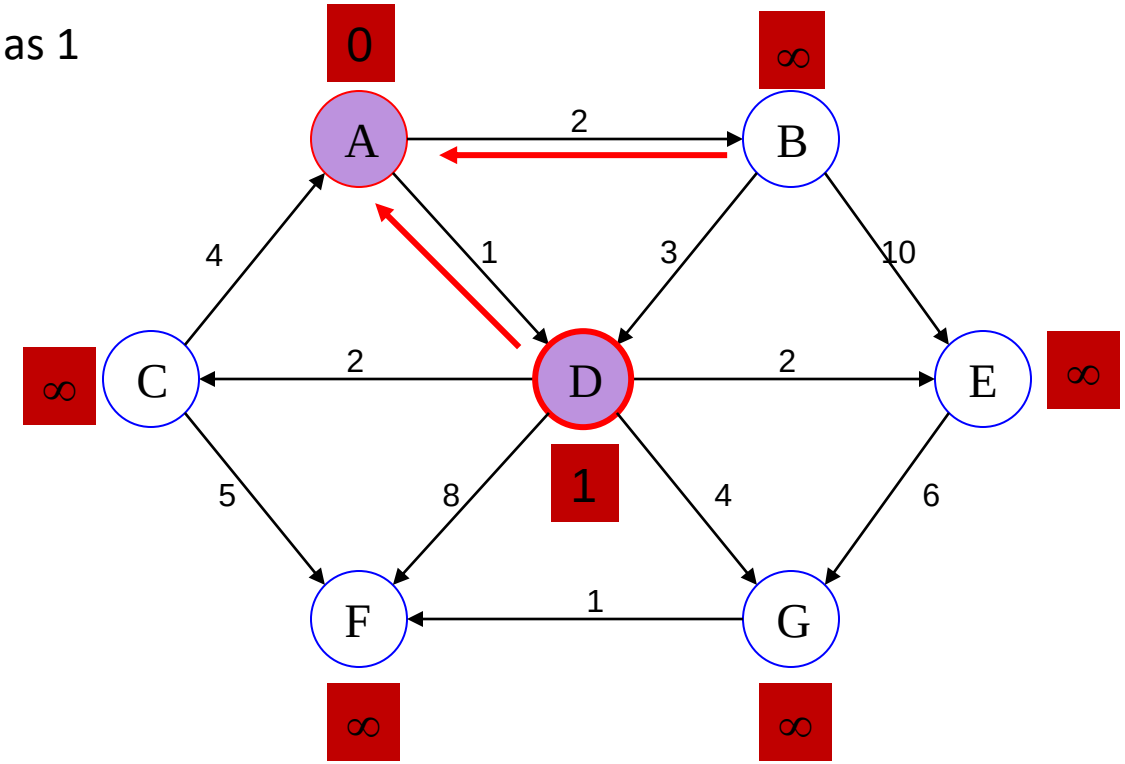


Distance (all vertices but source) = ∞

Pick vertex in List with minimum distance: D. (u) in the algorithm)

Remove the Vertex With Minimum Distance

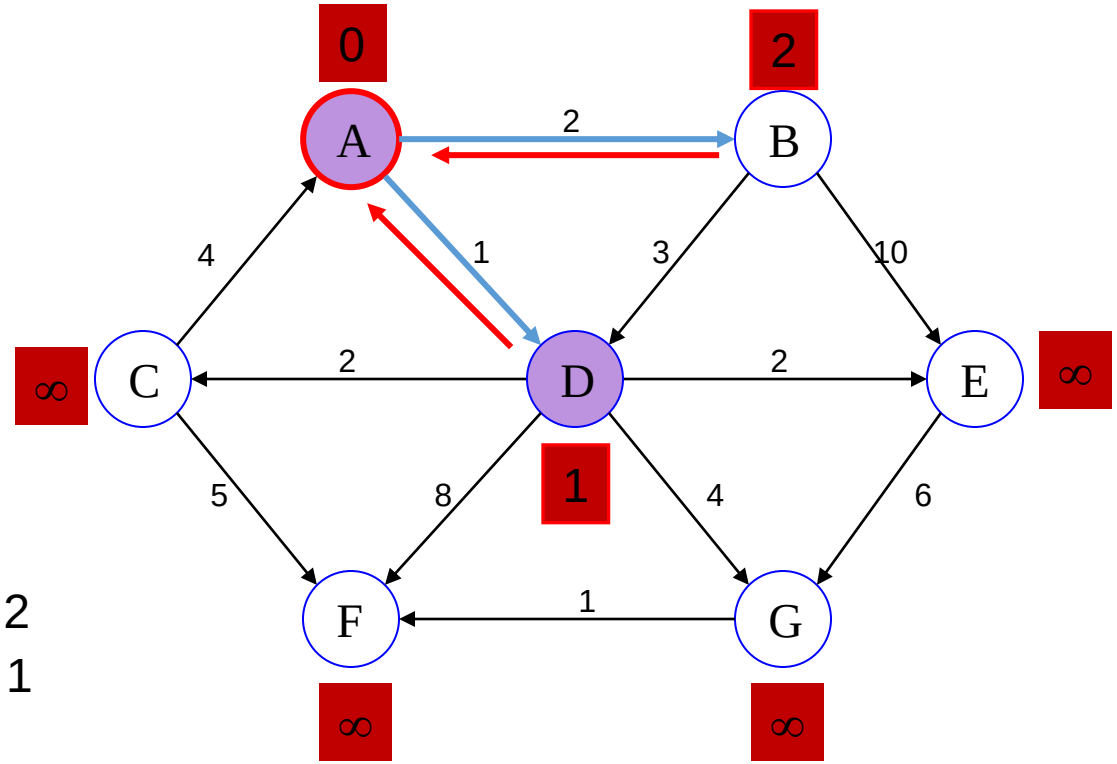
Update D's distance as 1



Mark D as **known. Add D: S= { A,D }**

Red arrows indicate paths from destination to source.

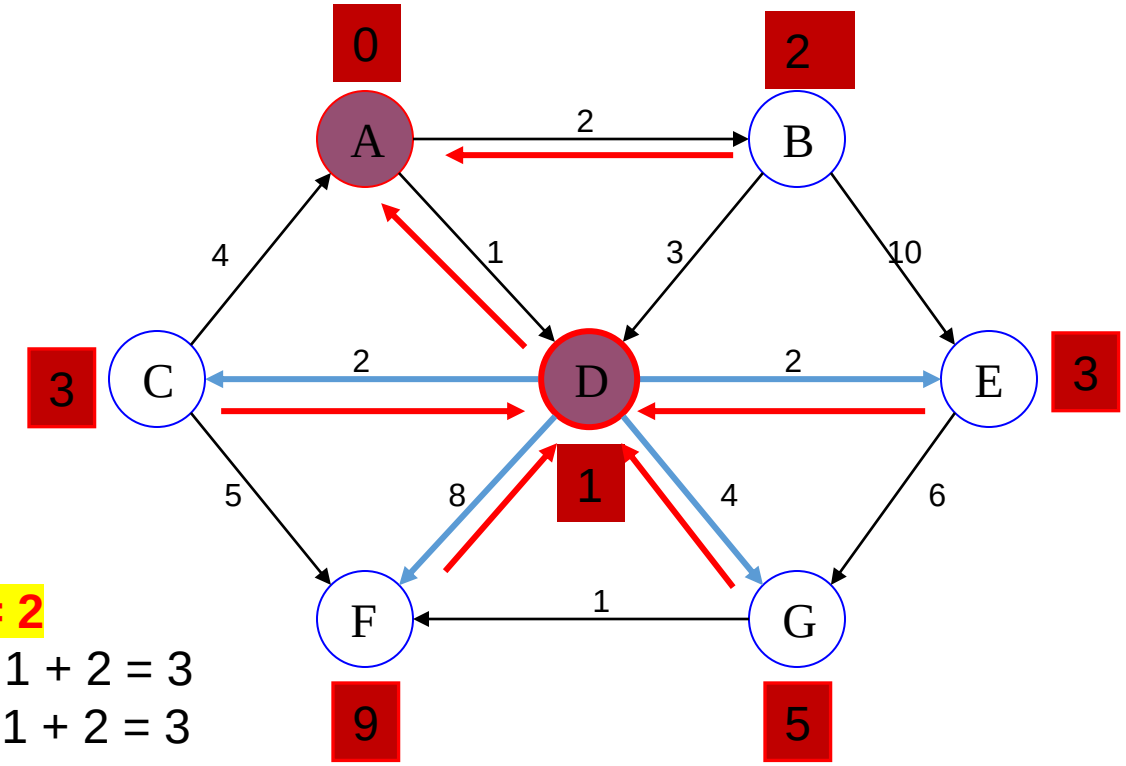
Update Neighbor Distances



Distance(B) = 2
Distance(D) = 1

Update neighbor distances:

Find all distances from A to others (Check one edge paths and all paths over D)



Distance(B) = 2

Distance(C) = 1 + 2 = 3

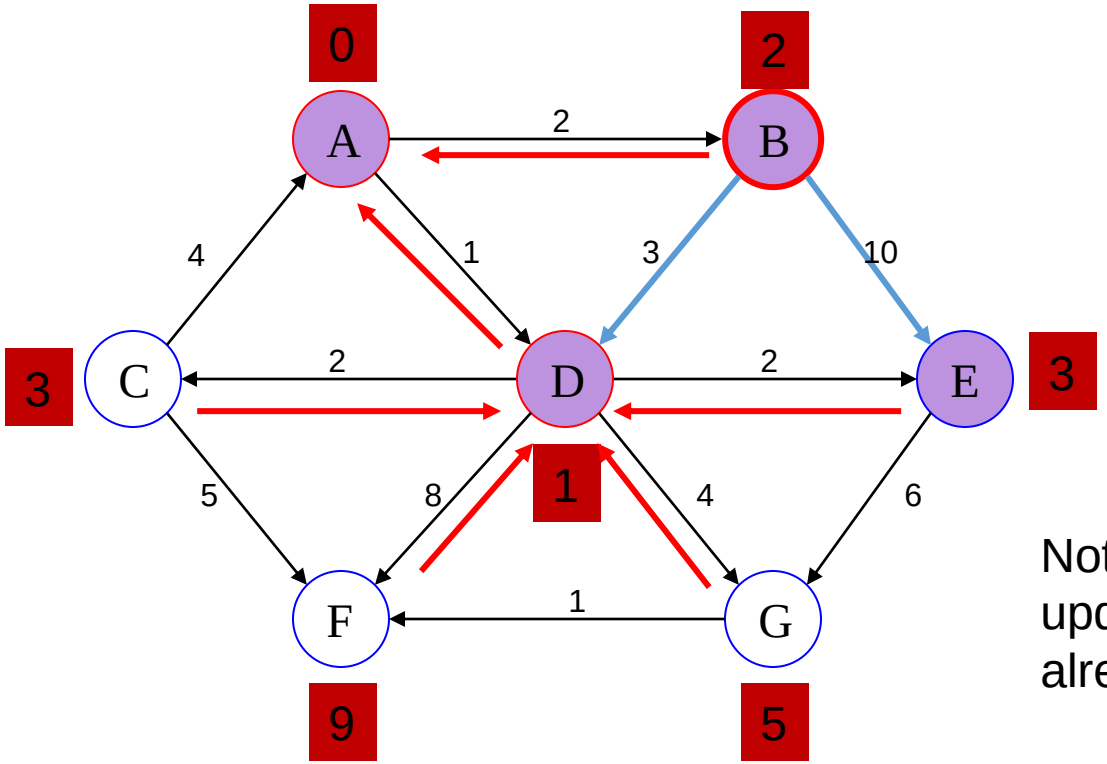
Distance(E) = 1 + 2 = 3

Distance(F) = 1 + 8 = 9

Distance(G) = 1 + 4 = 5

Update Neighbors

Pick vertex in the list with minimum distance: (B) . Update distances of B and E



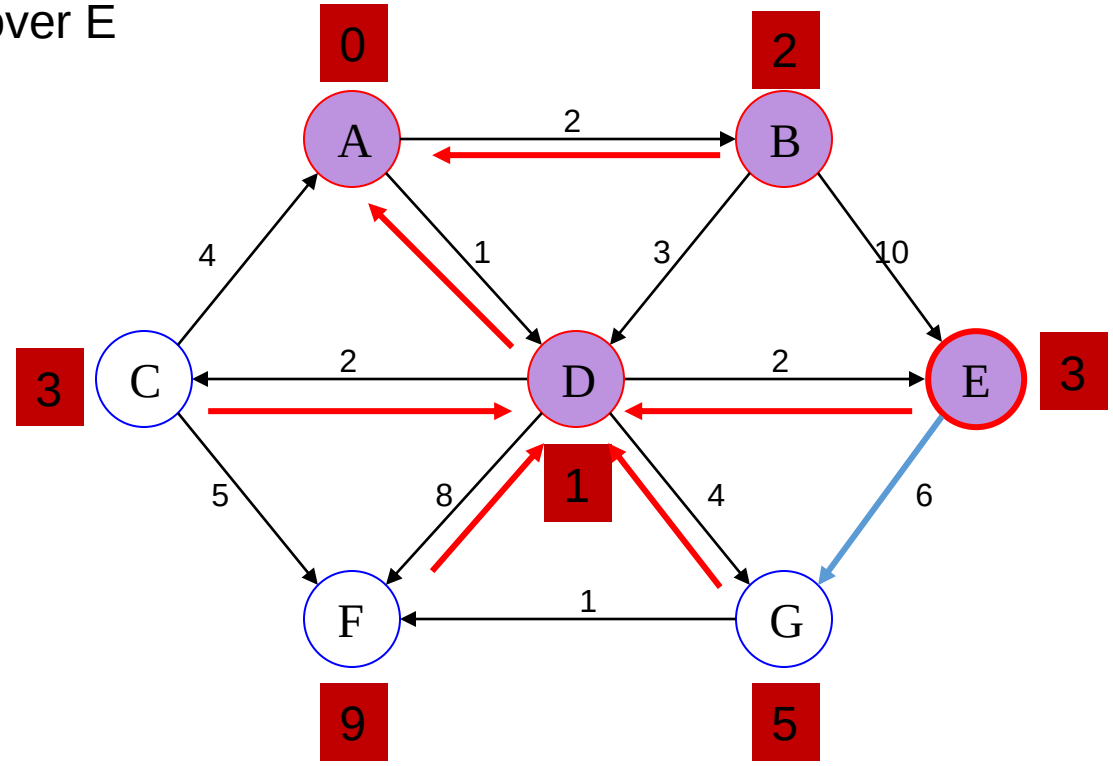
Note : distance(D) not updated since D is already known.

Mark B as known, S= {A,B,D}
Mark E as known , S={A,B,D,E}

Update Neighbors

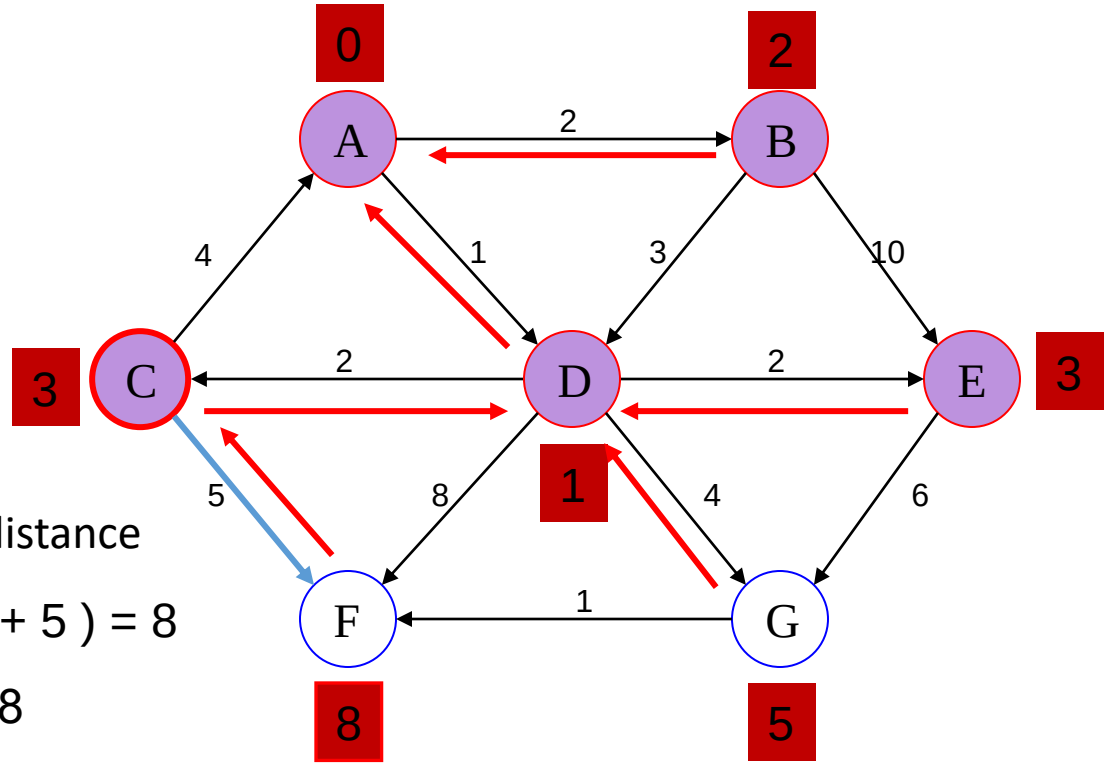
Pick List vertex over E with minimum distance: Update neighbors

No updating possible over E



Update Neighbors

Pick List vertex with minimum distance: C. Update neighbors



Previous distance

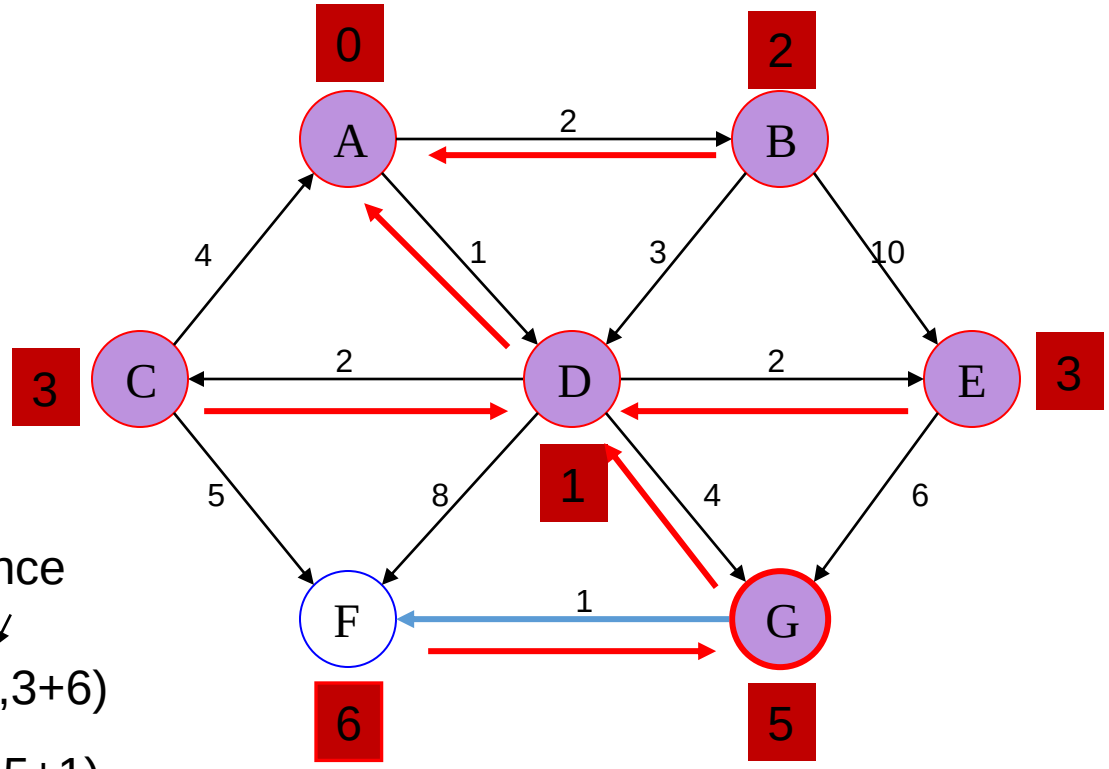
Distance(F) = min(9 , 3 + 5) = 8

Update F's distance as 8

Mark C as known. S={ A,B,C,D,E }

Update Neighbors

Pick vertex List with minimum distance : G. Update neighbors



Previous distance

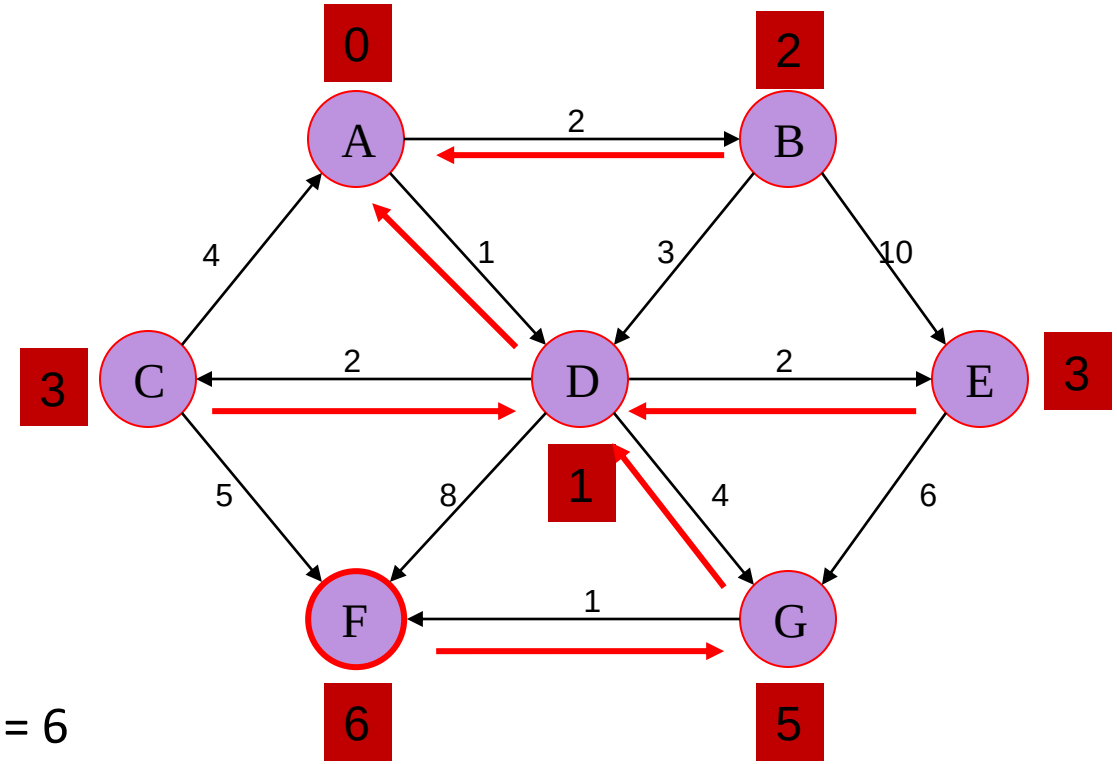
Distance(G) = min(5,3+6)

Distance(F)= min(8,5+1)

Mark G as known :S= {A,B,C,D,E,G}

Example Finished

Pick vertex not in S with lowest cost: F .Update neighbors. Last step.



Distance F = 6

Mark F as known. S= { A,B,C,D,E,F,G }

Minimal distances from A to all others have been determined(Numbers in squares)

Red arrows indicate paths from destination to source.

Time Complexity of Djkstra

Assume adjacency list implementation of graph G.

- We have |V| vertices and |E| edges. Costs in the algorithm:
  - Initialization: O(|V|)
  - While loop : O(|V|)
    - Nested for loop : Find and remove min distance vertices O(|V|)
    - Potentially |E| updates , each update cost: O(1)
- Reconstruct path O(|E|)

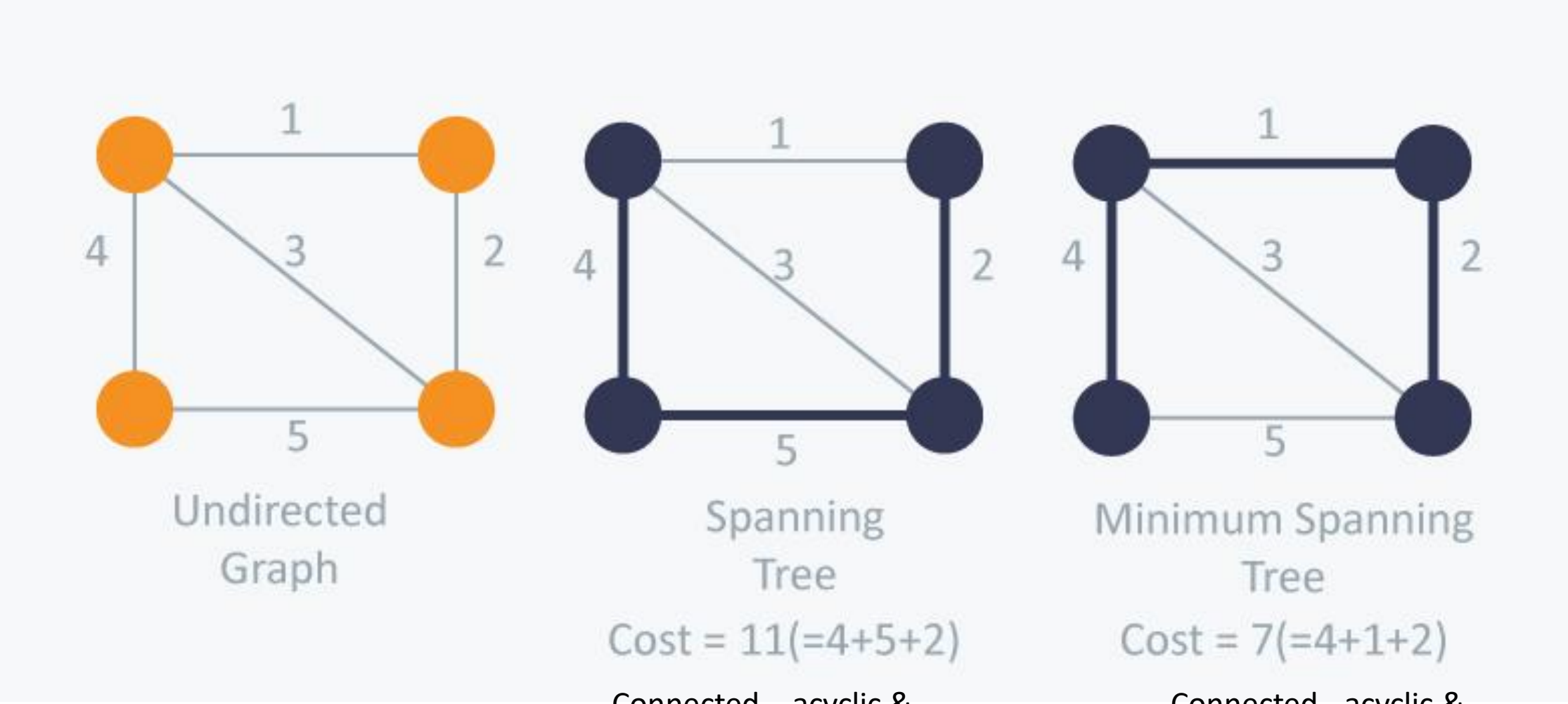
Total time T(n)= $(|V|^2 + |E|) = O(|V|^2)$

Minimum Spanning Trees

Tree→ a connected, directed acyclic graph

- Spanning tree: A subgraph of a graph, which meets the following constraints :
  - connected
  - acyclic
  - connects every vertex
- Minimum spanning tree(MST): A spanning tree with weight less than or equal to any other spanning tree for the given graph
- Two well-known algorithms for finding MST are Kruskal and Prim algorithms.

Minimum Spanning Tree : Example-1

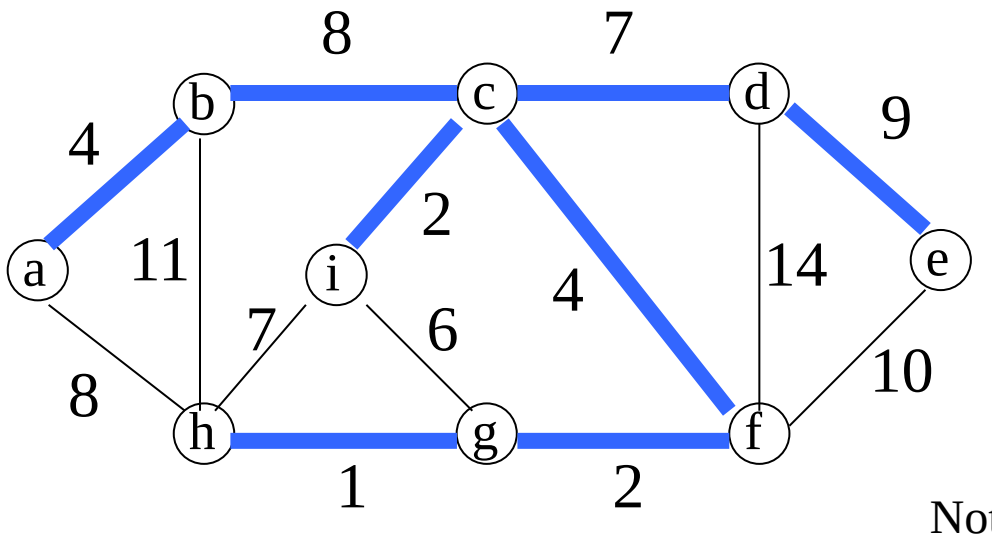


Connected – acyclic & connects every vertex

Connected - acyclic & connects every vertex minimum cost

Minimum Spanning Tree: Example-2

A connected graph and its MST:



Notice that the tree is not unique: replacing (b,c) with (a,h) yields another spanning tree with the same minimum weight.

How to find the MST?

Given a connected-weighted & undirected graph, find subset of edges that

- spans all the nodes

- creates no cycle,

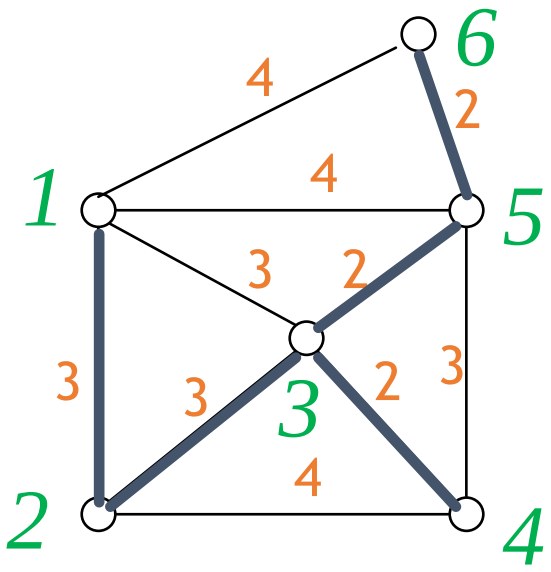
- minimizes the sum of weights.

Finding Minimum Spanning Tree: Greedy Solution

- Find a least-cost subset of the edges of a graph that connects all the nodes:
  - Start by picking any node and adding it to the tree
    - Repeatedly: Pick any *least-cost* edge from a node in the tree to a node not in the tree, and add the edge and new node to the tree
  - Stop when all nodes have been added to the tree
  - This is the Kruskal algorithm.

Finding Minimum spanning tree: Greedy Solution

Example: Start from node 1



- The result is a least-cost (3+3+2+2+2=12) spanning tree
- If you think some other edge should be in the spanning tree:
  - Try adding that edge
  - Note that the edge is part of a cycle
  - To break the cycle, you must remove the edge with the greatest cost
    - This will be the edge you just added

Generic MST Algorithm

input: weighted undirected graph

G = (V,E,w)

T = ø (empty set of edges)

while T is not yet a spanning tree of G

find an edge e in E such that T U {e} is a subgraph of some MST of G

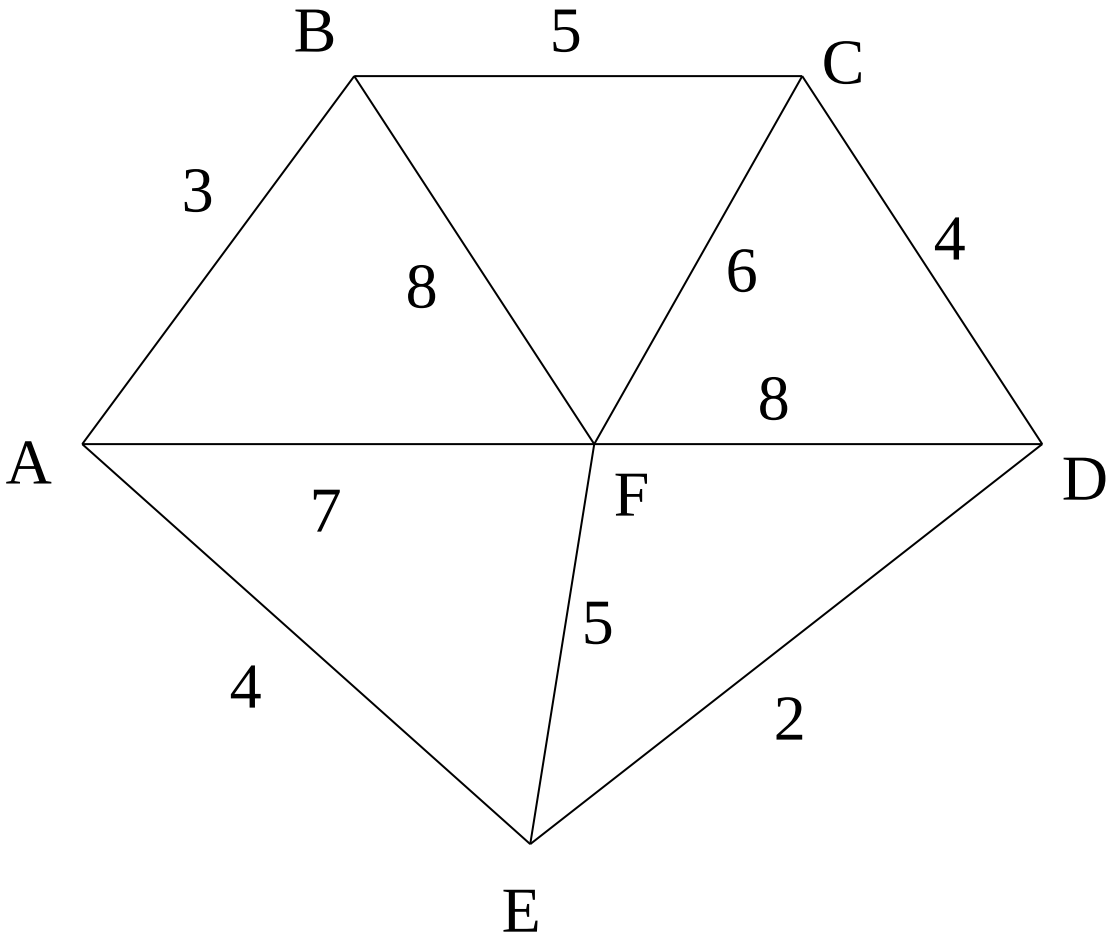
add e to T

return T //as MST of G

Kruskal's MST Algorithm

```
//G = (V, E,w)
Algorithm Kruskal ( G )
T ← ϕ        //The edges in MST,initially empty
Sort the m edges in G in increasing weight order
While | T |< |V| - 1 and E ≠ ϕ do
   Choose an edge (v, x) from E of lowest cost
   Delete (v, x) from E
   If (v, x) does not create a cycle in T
          then add (v, x) to T
           else discard (v, x)   //Unfeasible
Return T
```

Kruskal's Algorithm



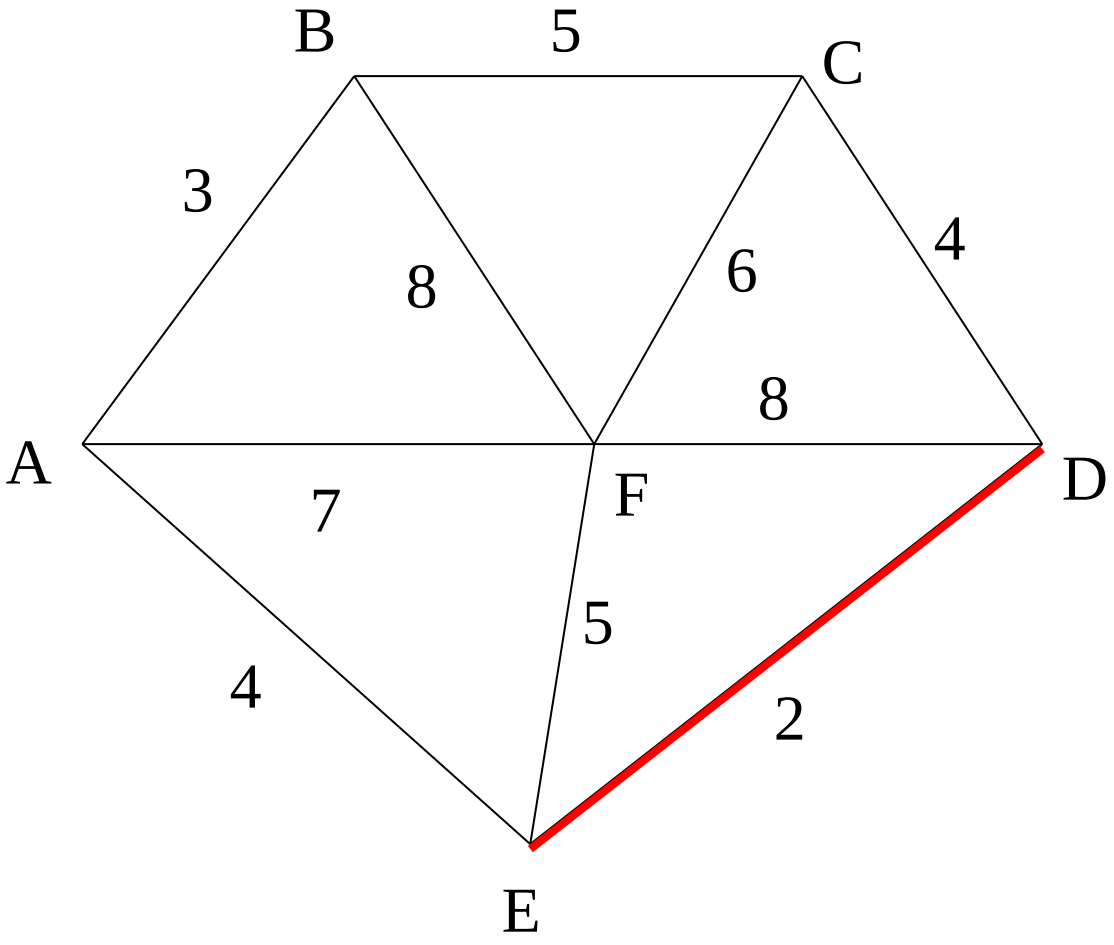
Sort the edges in order of size

ED 2
AB 3
AE 4
CD 4
BC 5
EF 5
CF 6
AF 7
BF 8
CF 8

Kruskal's Algorithm

Sorted List of the edges

ED 2
AB 3
AE 4
CD 4
BC 5
EF 5
CF 6
AF 7
BF 8
CF 8

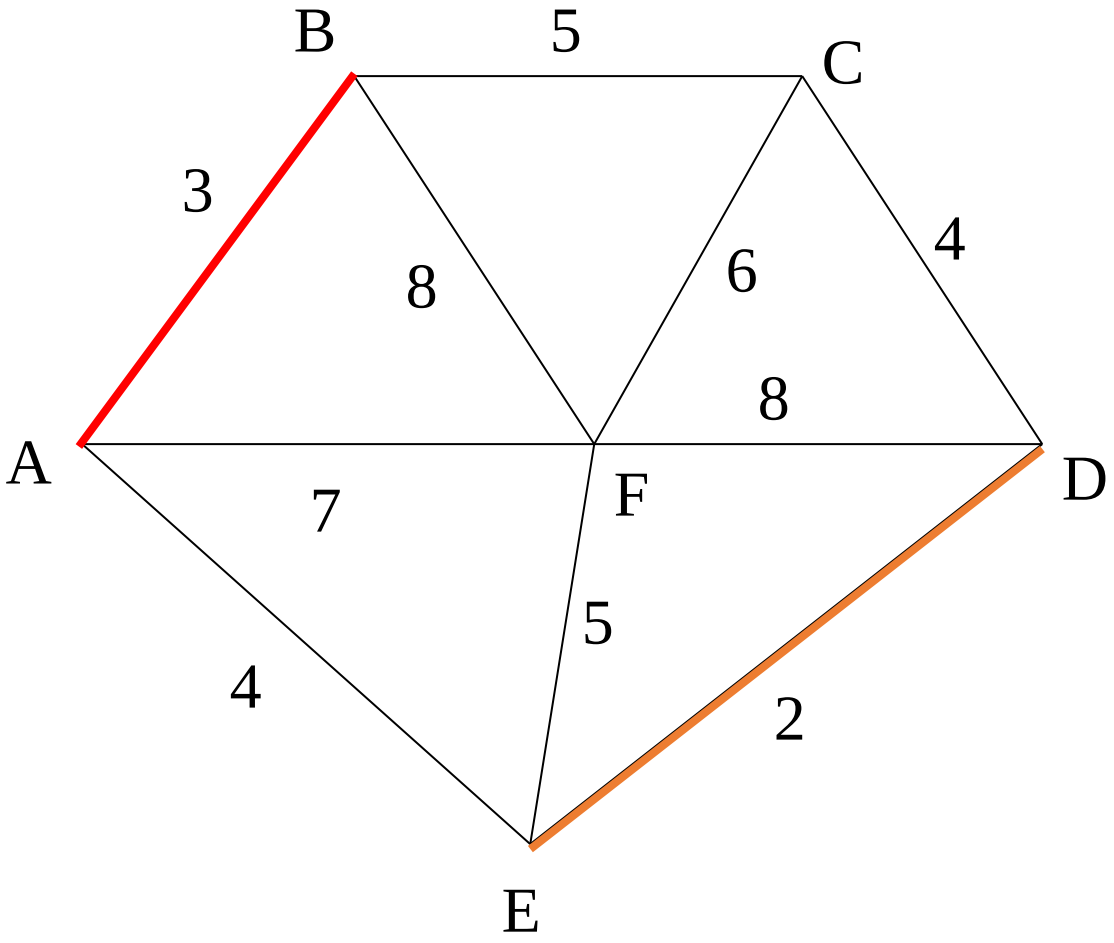


Select the shortest edge in the network :ED
No cycle,add to T:
ED 2

Kruskal's Algorithm

Sorted List of the edges

ED 2
AB 3
AE 4
CD 4
BC 5
EF 5
CF 6
AF 7
BF 8
CF 8



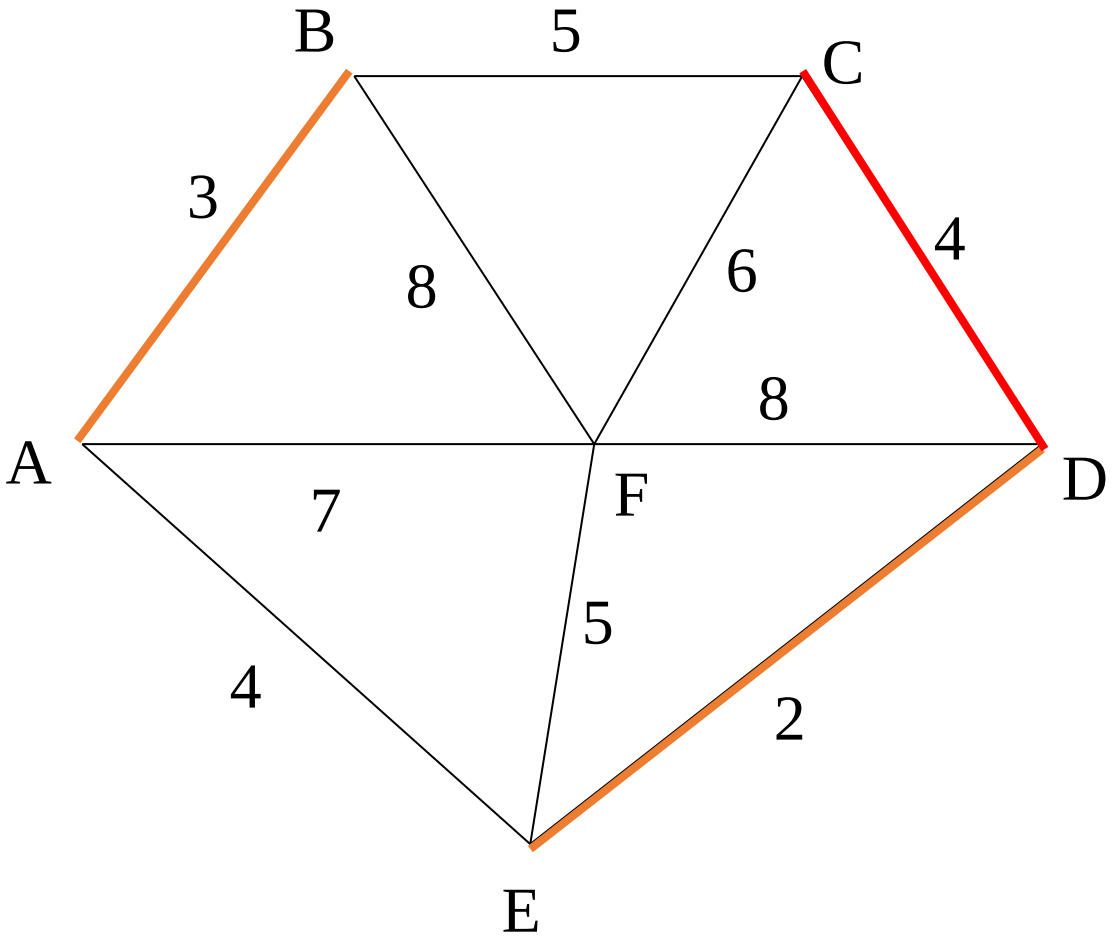
Select the next shortest edge which does not create a cycle: AB

ED 2
AB 3

Kruskal's Algorithm

Sorted List of the edges

ED 2
AB 3
AE 4
CD 4
BC 5
EF 5
CF 6
AF 7
BF 8
CF 8



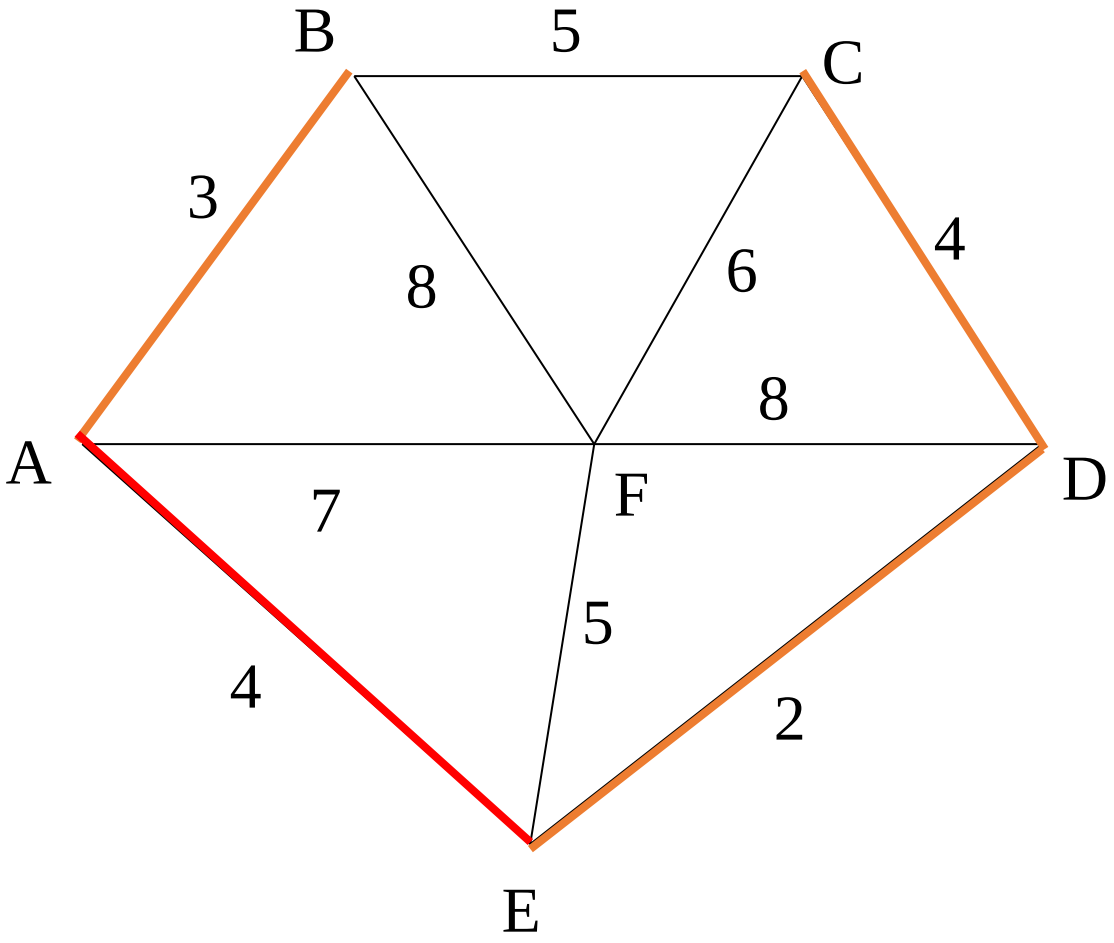
Select the next shortest edge which does not create a cycle :CD,AE

ED 2
AB 3
CD 4 (or AE 4)

Kruskal's Algorithm

Sorted List of the edges

ED 2
AB 3
AE 4
CD 4
BC 5
EF 5
CF 6
AF 7
BF 8
CF 8



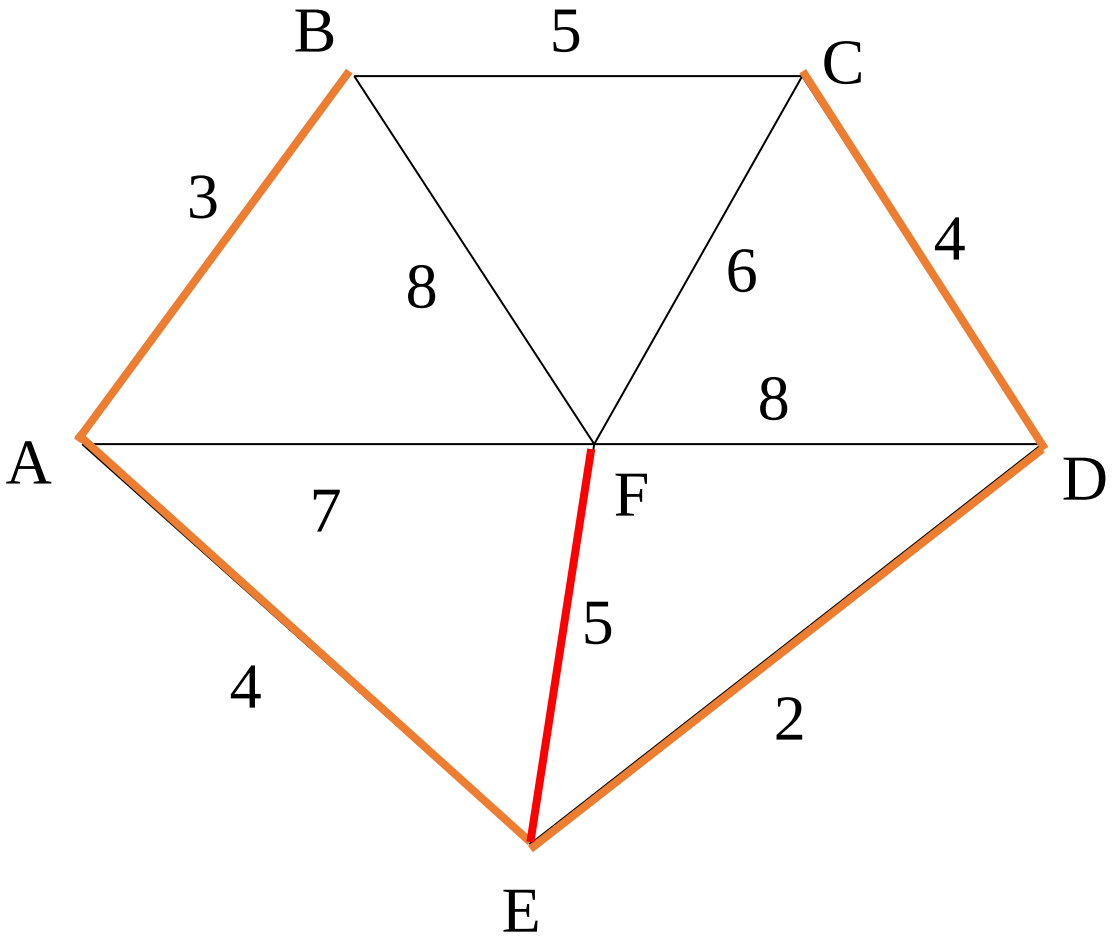
Select the next shortest edge which does not create a cycle

ED 2
AB 3
CD 4
AE 4

Kruskal's Algorithm

Sorted List of the edges

ED 2
AB 3
AE 4
CD 4
BC 5
EF 5
CF 6
AF 7
BF 8
CF 8



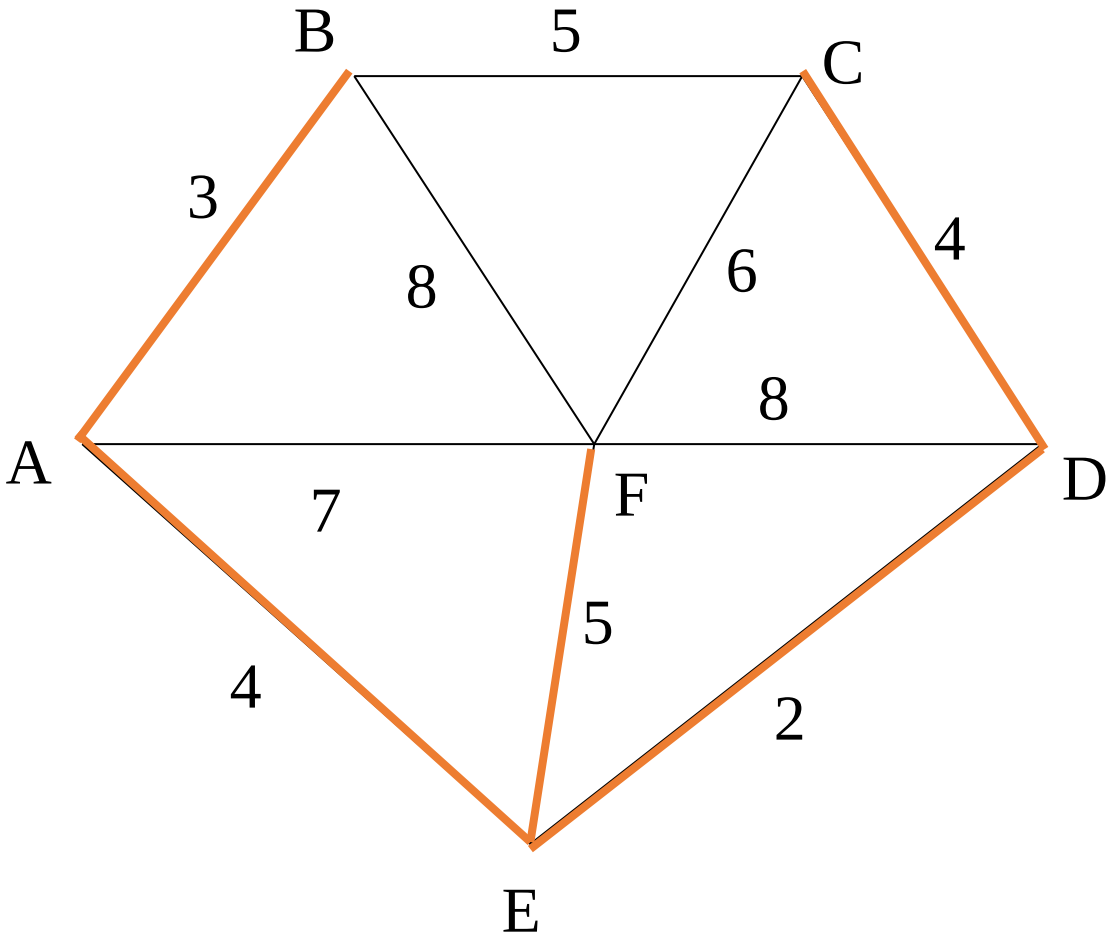
Select the next shortest edge which does not create a cycle BC,EF

ED 2
AB 3
CD 4
AE 4
BC 5 – forms a cycle
EF 5

Kruskal's Algorithm

Sorted List of the edges

ED 2
AB 3
AE 4
CD 4
BC 5
EF 5
CF 6
AF 7
BF 8
CF 8



All vertices have been connected.

The solution is

ED 2
AB 3
CD 4
AE 4
EF 5

Total weight of MST = 18

Why is Kruskal's Algorithm Greedy?

- Algorithm manages a set of edges such that
  - these edges are a subset of some MST
- At each iteration:
  - choose an edge so that the MST-subset property remains true
  - Sub problem has to do the same with the remaining edges
- Always try to add cheapest available edge that will not violate the tree property

  →locally optimal choice

Time Complexity of Kruskal

- Sorting of edges takes O(|E|Log |E|)time.
- After sorting, we iterate through all edges.
- The find and union operations can take at most O(Log|V|) time for each tree edge [This result is based on a priority queue implementation which will be considered later]

→So overall complexity is O(|E|Log |E | + |E|Log |V|) time.

- However, $|E| \sim |V|$, so we can write

$$\log(E) <= \log(|V|^2)$$

$$O(\log(E) = O(2 \log(|V|)) = O(\log(|V|))$$

so O(Log|V|) and O(Log|E|) are the same .

- Therefore, overall time complexity is

O(|E|log|E|) + O(|E|log|E|) = O(|E|log|E|)

Another Greedy MST Algorithm

- Kruskal's algorithm maintains a forest that grows until it forms a spanning tree
- Alternative idea is keep just one tree and grow it until it spans all the nodes:

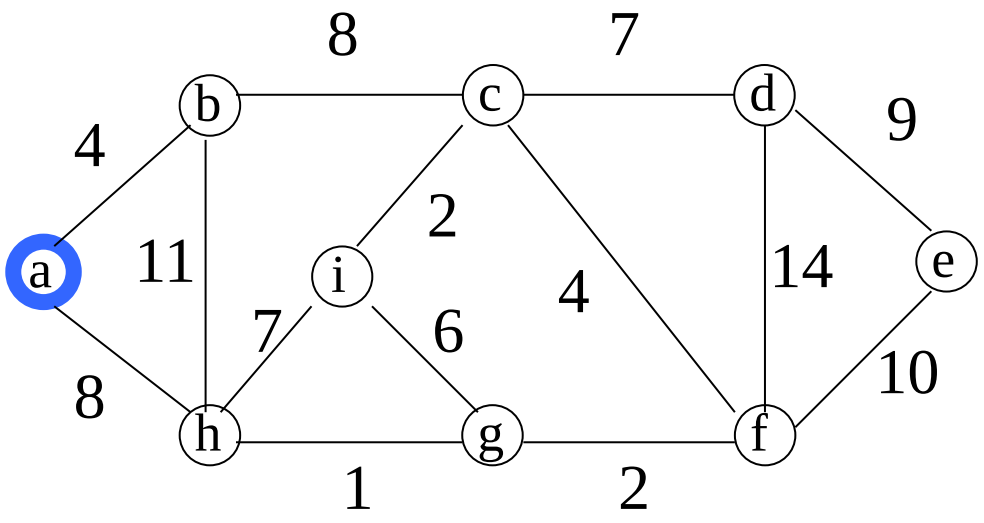
  Prim's algorithm
- At each iteration, choose the minimum weight outgoing edge : → Greedy!
- Problem: given a connected, undirected, weighted graph, find a *spanning tree* using edges that minimize the total weight

Prim's Algorithm

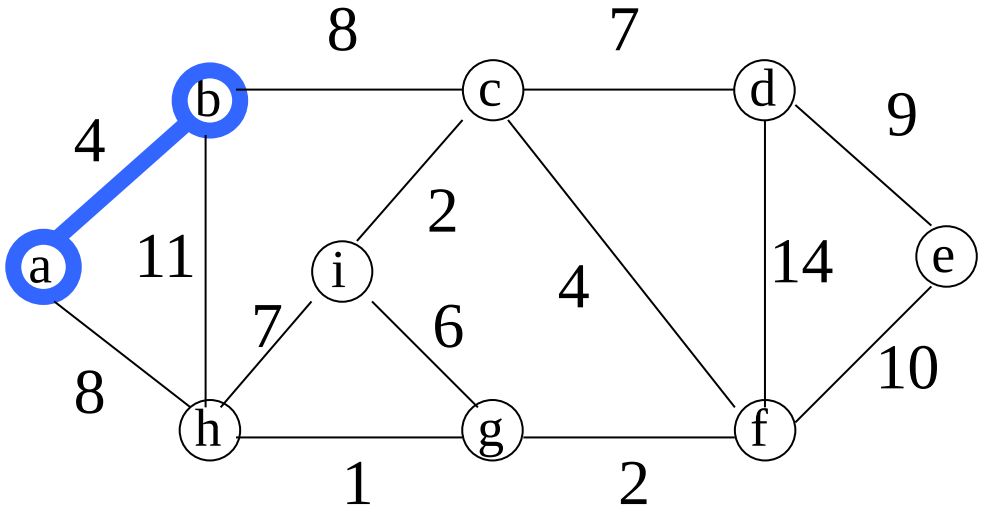
```
// input:  weighted undirected graph G = (V,E,w)
// r : Start node
MST_PRIM(G,r)
  T ← ϕ            //Edges in MST
  S←{r}            //The nodes in current tree
  Q←V-{r}    // Q: Remaning(outside) nodes
  while |T| < |V| - 1 do
     If  (u,v) is a min wt. outgoing edge  such that
          (u in S  and  v not in S)
     add (u,v) to T
     add v to S
     delete v from Q
return T
```

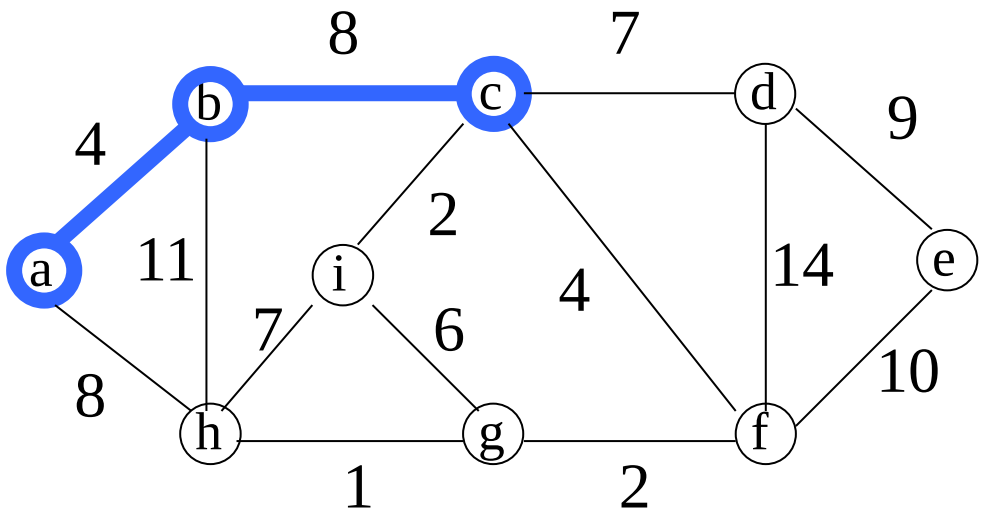
The execution of Prim's algorithm : Example

the root vertex: a



Min edge: (a,b)




8
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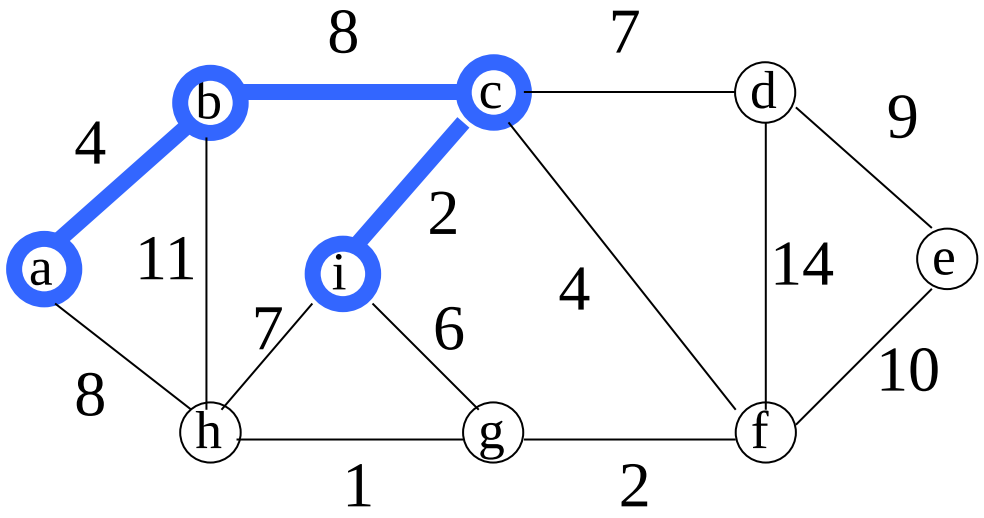
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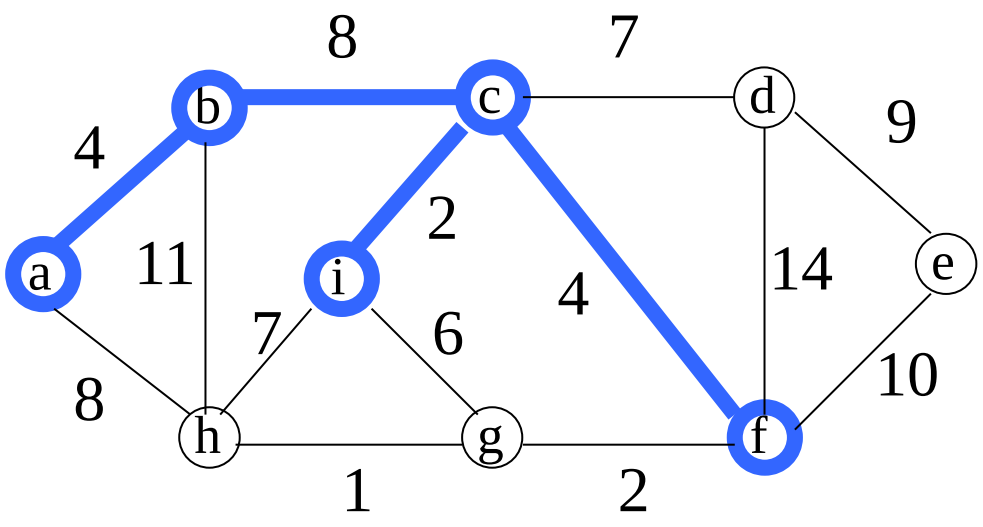
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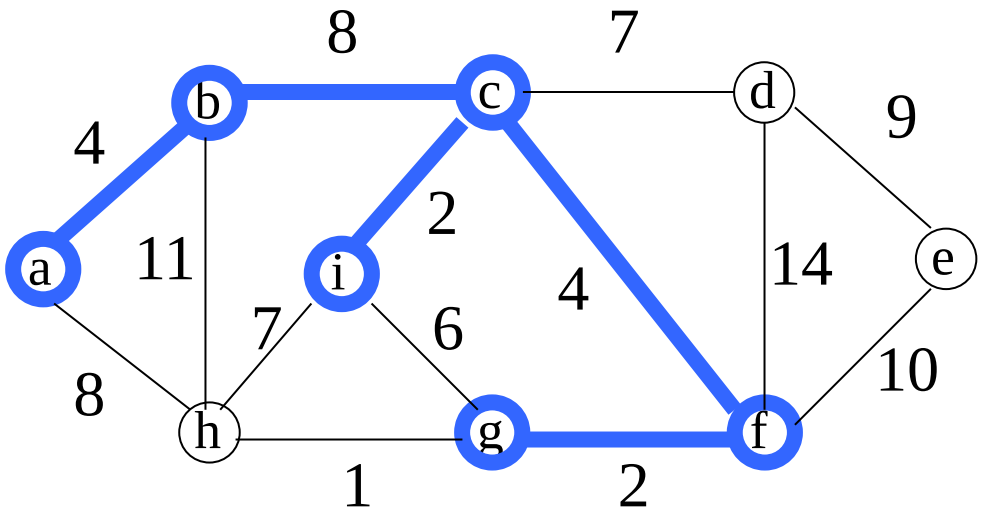
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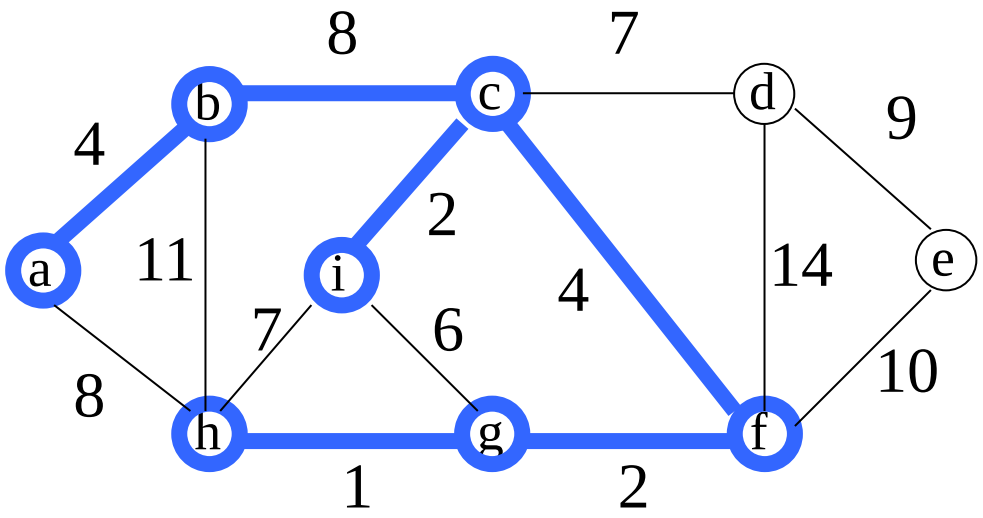
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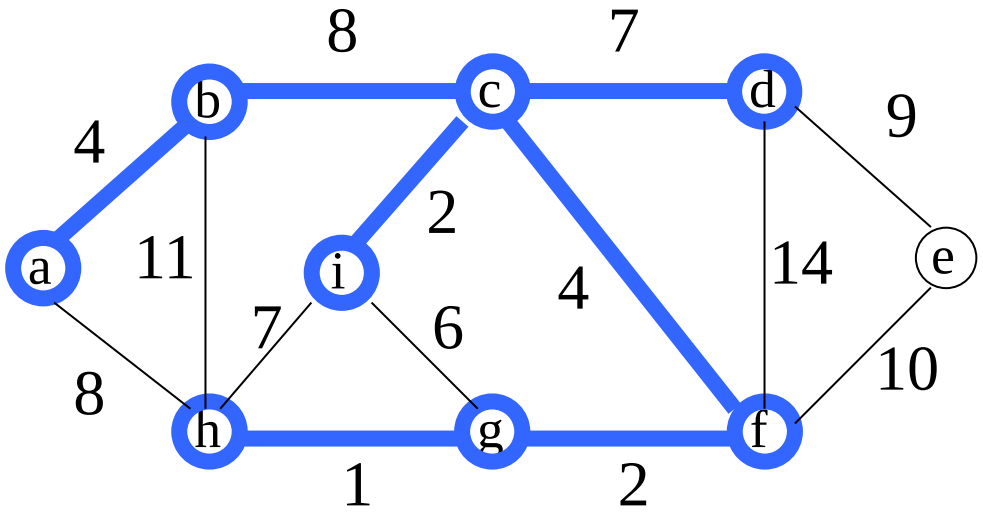
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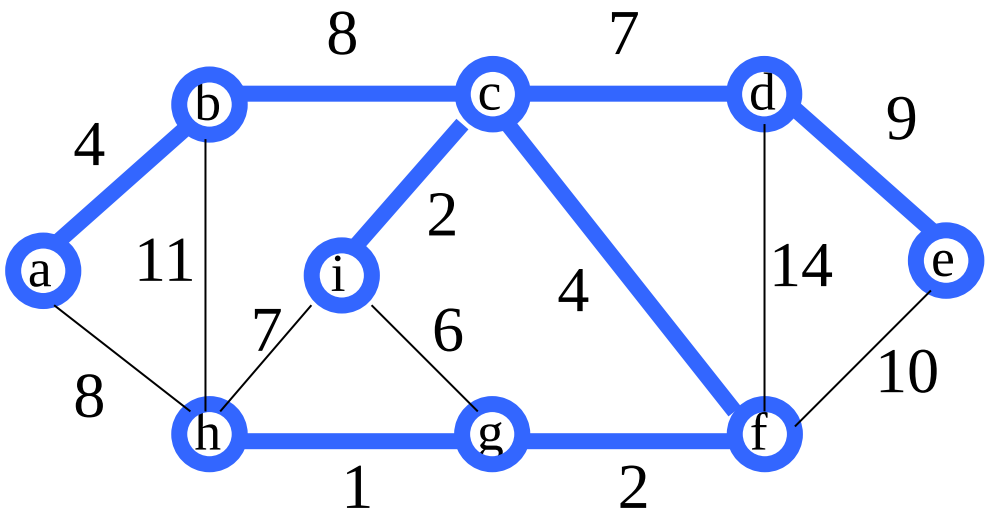
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Complexity of Prim's Algorithm

- Complexity depends on implementation. In our case:
- Initializations : O(|V|)
- While loop iteration: |V| times
- Each time we find a vertex, we must check all of its neighbors : O(|V|)
- Edge operations will be performed at most |E| times

With an adjacency list, complexity is:

$$O(|V|^2 + |E|) = O(2|V|^2) \quad (\text{ Using } |E| = O(|V|^2) \text{)}$$
$$= O(|V|^2 \text{)}$$

This complexity can be reduced using more efficient implementations involving priority queues.

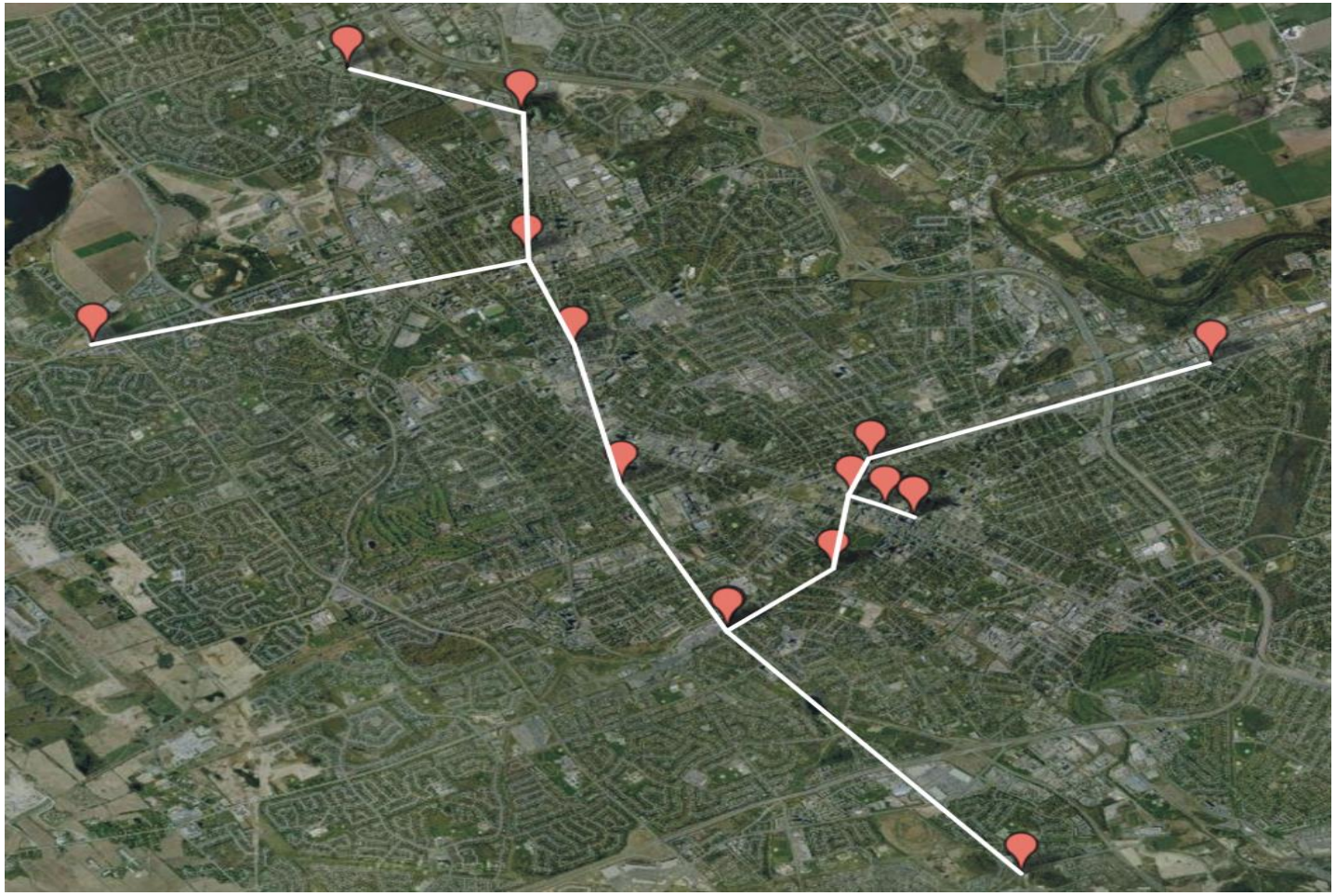
Applications of MST-1

Consider attempting to find the best means of connecting a number of LANs

- Minimize the number of bridges
- Costs not strictly dependant on distances

Applications of MST-1

A minimum spanning tree will provide the optimal solution:



Applications of MST-2

- In the design of electronic circuitry, it is often necessary to make a set of pins electrically equivalent by wiring them together.
- To interconnect n pins, we can use n-1 wires, each connecting two pins.
- We want to minimize the total length of the wires.
- Minimum Spanning Trees can be used to model this problem.

Applications of MST-3

- Consider a cable TV company laying cable to a new neighborhood...
  - If it is constrained to bury the cable only along certain paths, then there would be a graph representing which points are connected by those paths.
  - Some of those paths might be more expensive, because they are longer, or require the cable to be buried deeper.
    - These paths would be represented by edges with larger weights.
  - A spanning tree for that graph would be a subset of those paths that has no cycles but still connects to every house.
    - There might be several spanning trees possible. A minimum spanning tree would be one with the lowest total cost.

Prim's vs. Dijkstra's

- Both Prim’s and Kruskal’s Algorithms work with **undirected graphs**
- Both work with weighted and unweighted graphs but are more interesting **when edges are weighted**
- Both are greedy algorithms that produce optimal solutions